

Arbitrarily Delayed Continuation Splits in an Exact Lehmer-Signature Graph

Xiaojian Zeng*

September 7, 2026

Abstract

For the second iterated fixed layer of the Lehmer operator on inversion sequences, a natural exact right-extension state records a cumulative-count function together with the maximum and forbidden ranks of the current Lehmer image. We exhibit two reachable states at every level $m \geq 4$ whose labeled continuation languages agree through depth $m - 1$ and first separate at depth m . Every first-separating word is classified explicitly: it has the form $(y, 1^k, 0^{m-2-k}, 2)$. Thus no uniform finite lookahead distinguishes all right-language-inequivalent states of this literal signature graph. This is only an obstruction to one bounded-horizon state-minimization strategy. It does not give an efficient counting algorithm or a lower bound for arbitrary algorithms, and the motivating polynomial-bit-complexity counting problem remains unresolved.

1 Context and the exact extension graph

For $n \geq 0$, let

$$I_n = \{\sigma = (\sigma_1, \dots, \sigma_n) : 0 \leq \sigma_i < i\}$$

and define

$$\Theta(\sigma)_i = \#\{j < i : \sigma_j < \sigma_i\}.$$

Write $FP_n = \{\sigma \in I_n : \Theta(\sigma) = \sigma\}$ and

$$I_{n,2} = \{\sigma \in I_n : \overset{2}{\Theta}(\sigma) \in FP_n\}.$$

Allagan, Gao, and Testart introduced the iteration problem, characterized the first layer using avoidance of the patterns 101 and 201, and asked for efficient enumeration of higher layers [AGT26]. The motivating problem here is the stronger, explicit request for an exact polynomial-bit-complexity algorithm for $|I_{n,2}|$. We do not resolve that problem.

We first describe the particular exact state graph studied in this note. Let $\sigma \in I_{m,2}$ be a valid prefix, put $\tau = \Theta(\sigma)$, and define

$$F_\sigma(x) = \#\{j \leq m : \sigma_j < x\} \quad (0 \leq x \leq m), \quad M_\tau = \max_i \tau_i,$$

with $M_\tau = -1$ when $m = 0$. Set

$$B_\tau = \{r \in \mathbb{Z} : \text{some } a < b \leq m \text{ satisfy } \tau_a \geq r > \tau_b\}.$$

*PBC School of Finance, Tsinghua University, 100190, China. Email: xiaojianzeng@foxmail.com.

The exact signature of σ is

$$S(\sigma) = (F_\sigma, M_\tau, B_\tau).$$

The union of the patterns 101 and 201 is precisely the condition

$$q_a \geq q_c > q_b \quad (a < b < c). \quad (1)$$

Consequently, the cited pattern criterion says that $\sigma \in I_{m,2}$ if and only if τ avoids (1).

If $x \in \{0, \dots, m\}$ is appended to σ , then the new entry of its Lehmer image is $r = F_\sigma(x)$. Since the old image already avoids (1), the extension is valid exactly when $r \notin B_\tau$. When it is valid, its successor signature is determined by the current signature and the label x :

$$\begin{aligned} F_{(\sigma,x)}(z) &= F_\sigma(z) + \mathbf{1}_{\{x < z\}} & (0 \leq z \leq m), \\ F_{(\sigma,x)}(m+1) &= m+1, \\ M_{(\tau,r)} &= \max(M_\tau, r), \\ B_{(\tau,r)} &= B_\tau \cup \{q \in \mathbb{Z} : r < q \leq M_\tau\}. \end{aligned}$$

Thus the signatures form a deterministic labeled right-extension graph; we call a state reachable when it is the signature of an element of some $I_{m,2}$.

For a state U at level m , an append word $w = (x_1, \dots, x_t)$ has the inversion-sequence bounds $0 \leq x_i \leq m + i - 1$. It is *valid from* U when all of its labeled transitions exist. Let $\mathcal{L}_t(U)$ be the set of valid length- t words. We also use the weighted, label-forgetting continuation profiles

$$H_0(U) = *, \quad H_{d+1}(U) = \text{multiset}_{x:U_x \text{ valid}} H_d(U_x), \quad (2)$$

where distinct labels are retained with multiplicity.

2 The delayed-split theorem

For repeated entries we use exponential notation, and for $m \geq 4$ put

$$\alpha_m = (0, 1, 2^{m-2}), \quad \beta_m = (0, 0, 1, 2^{m-3}).$$

Their Lehmer images are, respectively,

$$(0, 1, 2^{m-2}) \quad \text{and} \quad (0, 0, 2, 3^{m-3}).$$

Both are nondecreasing, so both roots are reachable in $I_{m,2}$. Directly, their signatures are

$$\begin{aligned} S_m &= ((0, 1, 2, m^{m-2}), 2, \emptyset), \\ T_m &= ((0, 2, 3, m^{m-2}), 3, \emptyset). \end{aligned} \quad (3)$$

Theorem 1. *For every integer $m \geq 4$,*

$$\mathcal{L}_t(S_m) = \mathcal{L}_t(T_m) \quad (0 \leq t < m).$$

At length m , the symmetric difference is exactly

$$\mathcal{L}_m(T_m) \setminus \mathcal{L}_m(S_m) = \{(y, 1^k, 0^{m-2-k}, 2) : 3 \leq y \leq m, 0 \leq k \leq m-2\}, \quad (4)$$

and $\mathcal{L}_m(S_m) \setminus \mathcal{L}_m(T_m)$ is empty. Hence

$$H_{m-1}(S_m) = H_{m-1}(T_m), \quad H_m(S_m) \neq H_m(T_m).$$

In particular, no fixed finite continuation horizon distinguishes all pairs of right-language-inequivalent reachable signatures in this exact graph.

Proof. Fix $m \geq 4$ and a common append word $w = (x_1, \dots, x_t)$. Appending the same word adds the same multiset to the two underlying inversion sequences. Comparison of the two root multisets therefore gives, for every integer x ,

$$F_{\beta_m, w}(x) - F_{\alpha_m, w}(x) = \mathbf{1}_{\{1, 2\}}(x). \quad (5)$$

For $i \geq 1$, write

$$N_i(x) = \#\{j < i : x_j < x\}.$$

The ranks appended to $\Theta(\alpha_m)$ and $\Theta(\beta_m)$ are r_i and s_i , where

$$r_i = \begin{cases} 0, & x_i = 0, \\ 1 + N_i(1), & x_i = 1, \\ 2 + N_i(2), & x_i = 2, \\ m + N_i(x_i), & x_i \geq 3, \end{cases} \quad s_i = r_i + \mathbf{1}_{\{1, 2\}}(x_i). \quad (6)$$

Call ranks coming from symbols 1 or 2 *marked*, those from 0 *zero*, and those from symbols at least 3 *high*. Thus only marked ranks increase, by exactly one, from the first root to the second.

Each initial Lehmer image is nondecreasing and ends at its maximum. In a triple (1), the middle index cannot lie in the initial block. When only the first index lies there, its entry may be replaced by the final initial maximum. It follows that validity of w from the two roots is equivalent to avoidance of (1) by the two virtual sequences

$$A(w) = (2, r_1, \dots, r_t), \quad C(w) = (3, s_1, \dots, s_t). \quad (7)$$

For the comparison, regard the leading entries in (7) as marked as well.

We first show, for $t \leq m$, that if $C(w)$ contains (1), then $A(w)$ does. Take a witnessing triple and denote its closing word position by c . A zero rank cannot close the triple. If the closing rank is high, say $h = m + N_c(x_c)$, every earlier marked rank in $A(w)$ is at most $N_c(x_c) + 2 \leq h - 2$, and hence its counterpart in $C(w)$ is at most $h - 1$. The first entry of the witness must therefore be high, and decreasing marked entries by one preserves the witness.

Suppose instead that the closing rank is marked. If the middle rank is marked, it decreases together with the closing rank; if it is zero, the strict lower inequality is again preserved. If the middle rank is high, the lower inequality could fail in $A(w)$ only at the boundary $r_c = h$. The upper inequality in $C(w)$ would then require an earlier rank at least $h + 1$. The virtual leading rank is $3 \leq h$, and an earlier marked rank at word position $i \leq c - 2$ is at most $s_i \leq i + 2 \leq c \leq m \leq h$. If the required first rank were high, there would be two high symbols before c , forcing $r_c \leq c - 1 \leq m - 1 < h$. Both alternatives are impossible. Finally, the upper inequality transfers because a marked first rank decreases together with the closing rank, whereas a high first rank is unchanged. This proves the implication.

Now suppose that $A(w)$ contains (1) while $C(w)$ avoids it, still with $t \leq m$. A zero rank cannot close a witness, and the preceding high-rank bound shows that a witness closing at a high rank would survive in $C(w)$. Thus its closing rank is marked. Its middle entry cannot be high: the presence of a preceding high symbol gives $r_c \leq c \leq m$, whereas every high rank is at least m , and the lower inequality is strict. The lower inequality consequently survives in $C(w)$. Failure of the upper inequality is possible only when the first entry is an unshifted high rank exactly equal to r_c .

If $x_c = 1$, the preceding high symbol gives $r_c \leq c - 1 < m$, which cannot equal a high rank. Hence $x_c = 2$, and (6) yields

$$m \leq r_c = 2 + N_c(2) \leq c \leq t \leq m. \quad (8)$$

Equality holds throughout. Thus $c = t = m$; among the preceding symbols there is exactly one high symbol and all remaining $m - 2$ symbols are 0 or 1. The high rank equals m , so nothing precedes that high symbol; it is the first word symbol, say y , with $3 \leq y \leq m$. We have

$$w = (y, u, 2), \quad u \in \{0, 1\}^{m-2}. \quad (9)$$

If a 0 in u preceded a later 1, the high rank m , that zero rank, and the later marked rank would form (1) in $C(w)$. Since $C(w)$ avoids (1), all the 1s precede all the 0s, so $u = 1^k 0^{m-2-k}$.

Conversely, every word in (4) gives exactly

$$A(w) = (2, m, 1^k, 0^{m-2-k}, m), \quad C(w) = (3, m, 2^k, 0^{m-2-k}, m+1). \quad (10)$$

The first sequence contains (1), using its first m , any intervening low rank, and its final m . Before its final entry the second sequence never rebounds after a descent, and its final entry is a new maximum, so it avoids (1). Its corresponding word is therefore valid from T_m and invalid from S_m ; all proper prefixes are valid from both. This proves (4), and the earlier comparison proves equality at every smaller length.

The labeled continuation trees are identical through depth $m - 1$, which implies equality of the profiles in (2). At depth m , the second tree has exactly $(m - 1)(m - 2)$ additional leaves, so the profiles differ. Given any fixed horizon d , choosing $m > d$ produces two states with equal depth- d profiles but different full right languages. This proves the final claim. $\square$

3 Exact falsification checks and scope

An exact coupled traversal of the two signature roots was used to attack the statement before the proof was accepted. For $m = 4, 5, 6, 7, 8$, it found no split below depth m and found respectively

$$6, 12, 20, 30, 42$$

first-divergent words at depth m , exactly the words in (4). The traversal uses integer state transitions, with neither randomness nor floating-point arithmetic. A separately written implementation starting from the definition of Θ exhaustively checked every admissible word and prefix for $m = 4, 5, 6$. These finite computations are corroboration and falsification attempts, not premises of Theorem 1. Exact finite coefficient verification also appears as a certification method in a different enumerative setting in [Zen26]; we cite it only for its distinct-cycle-length coefficient certification, not as a source on inversion sequences or Lehmer signatures.

The theorem concerns only the literal (F, M, B) graph and a uniform fixed lookahead. It does not exclude a horizon growing with the level, a different symbolic quotient, an algebraic generating function, or any other polynomial-time counting method. It gives neither a superpolynomial state lower bound nor an algorithm for $|I_{n,2}|$. Its contribution relative to the motivating source is an infinite reachable family with an arbitrarily late first continuation split and an exact classification of all first-divergent words. No absolute priority claim is made beyond the bounded source comparison described here.

Computational and AI assistance. The finite searches were performed with exact Python programs. OpenAI Codex assisted with exploration, proof checking, and manuscript preparation. The all-parameter conclusion rests on the displayed mathematical argument; the computer calculations were used only to discover examples and to seek counterexamples.

References

- [AGT26] Julian Allagan, Shanzhen Gao, and Benjamin Testart. Iterating the Lehmer code on inversion sequences: Catalan fixed points and finite stabilization, 2026. arXiv:2608.24476 [math.CO].
- [Zen26] Zijian Zeng. The distinct-cycle-length probability is strictly decreasing after 30, 2026. doi:10.2139/ssrn.7380519.